



Assignment 1

Robotic challenge solver using the CiberRato
simulation environment

Project Objectives

01

C1 – Control Challenge

Navigate a closed circuit, avoiding collisions and completing as many laps as possible.

02

C2 – Mapping Challenge

Explore an unknown maze and create an accurate map of the environment.

03

C3 – Planning Challenge

Locate targets in the maze and plan the shortest path that visits all of them.

01

C1 – Control Challenge

Navigate a closed circuit as quickly as possible
without colliding with walls.

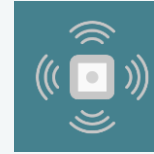


C1 – Strategy



PID Control Algorithm

- 🔧 Used to keep the robot centered in the path by adjusting based on distance errors from the walls.
- 🔧 Tuned constants (K_p , K_i , K_d) to balance responsiveness and stability, avoiding sharp turns.



Sensor Filtering

- 🔧 Applied a moving average filter to smooth sensor data, reducing noise and improving accuracy.

C1 – Navigation and Collision Handling

🔍 Obstacle Detection

Monitors front, left, and right sensors to detect nearby walls.

🔍 Directional Adjustments

PID-driven adjustments to motor speeds keep the robot on course.

🔍 Collision Recovery

Automatically reverses and turns if a collision is detected, allowing for quick recovery.

02

C2 – Mapping Challenge

Explore an unknown maze and generate an accurate map of the environment.



C2 – Strategy



Positioning with GPS & Compass

Used to continuously track the robot's location and orientation within the maze.



Wall Detection and Mapping

Side sensors identify nearby walls, and data is stored in a grid map, updating cells as walls or open spaces.

C2 – Navigation and Mapping Algorithms

🔧 PID for Movement

PID controllers maintain precise alignment and speed as the robot moves within cells.

🔧 Mapping Function

Records walls and free paths around the robot based on its orientation.

🔧 Pathfinding with BFS

Breadth-First Search (BFS) is used to find the shortest route to unexplored areas, allowing for efficient maze coverage.

Conclusion

Key Takeaways

- PID control, filtering sensor data, and navigation algorithms for robotics.

Project Outcomes

- **C1:** Improved control accuracy, allowing the robot to complete multiple laps without collisions.
- **C2:** Successfully generated a detailed map of the maze, identifying walls and open paths.

Future Improvements

- Optimize PID constants for even smoother control.
- Enhance exploration algorithms for more efficient and complete mapping.

Thanks!

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